Lecture 11

2/13/26

Phase Contrast (continued)

Phase contrast μscopy is used to make otherwise invisible specimen-induced phase shifts into visible amplitude shifts

PC is commonly used w/ tissue-culture μscopes
- used to monitor cell growth
- works w/ plastic substrates (e.g., w/ Petri dishes)

PC is a Nobel Prize winning technique

Looked again at images comparing BF & PC (see PowerPoint slides)

Implementation of Phase Contrast

How do you implement phase contrast? Understand w/ vector view

Z = zeroth-order light
D = diffracted light
T = total light

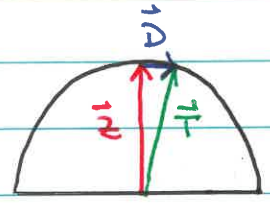
$$\boxed{\vec{T} = \vec{Z} + \vec{D}}$$

$\vec{Z}$ • Most light passes thru specimen unaffected

$\vec{D}$ • Small amount of light is affected

$\vec{T}$ • Total light is sum of unaffected & affected light

Phase objects, without some "trick," are not visible in BF μscopy because they do <u>NOT</u> absorb light

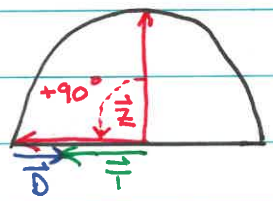


$|\vec{T}| = |\vec{Z}|$ • Uniform Kohler background illumination

$\vec{D} \perp \vec{Z}$ • No specimen information

Zernike realized that if you can create interference you can see light contrast

Try to advance the $\vec{Z}$ wave by 90° so it can interfere w/ the specimen-dependent $\vec{D}$ wave



$|\vec{T}| < |\vec{Z}|$

$\vec{D}$ is unaltered

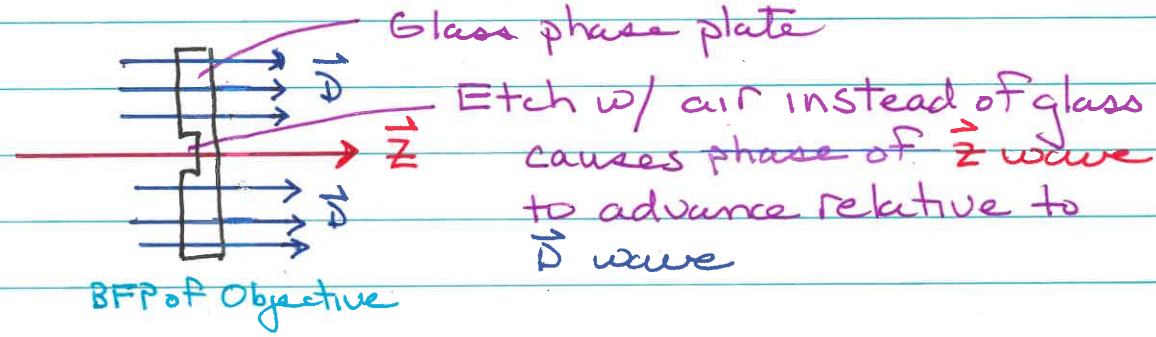
This is termed "positive" phase contrast because the phase of $\vec{Z}$ is advanced

- Background = $\vec{Z}$ alone ⇒ uniform w/o cells
- Specimen = $\vec{T} < \vec{Z}$ ⇒ specimen darker than backgrd
- Specimen signal $\vec{T}$ varies w/ position because different points in cell diffract differently
- Get point-to-point contrast in $\vec{T}$

Implementation Pt 1 – Point light source

The $\vec{z}$ wave can be phase advanced relative to the $\vec{D}$ wave by sending both waves through a glass phase plate but sending the $\vec{z}$ wave through a thinner piece of glass than the $\vec{D}$ wave

This is done in the BFP of the objective



$\vec{z}$ wave will "get ahead" traveling thru air instead of glass, advancing its phase, because light travels faster in air than in glass

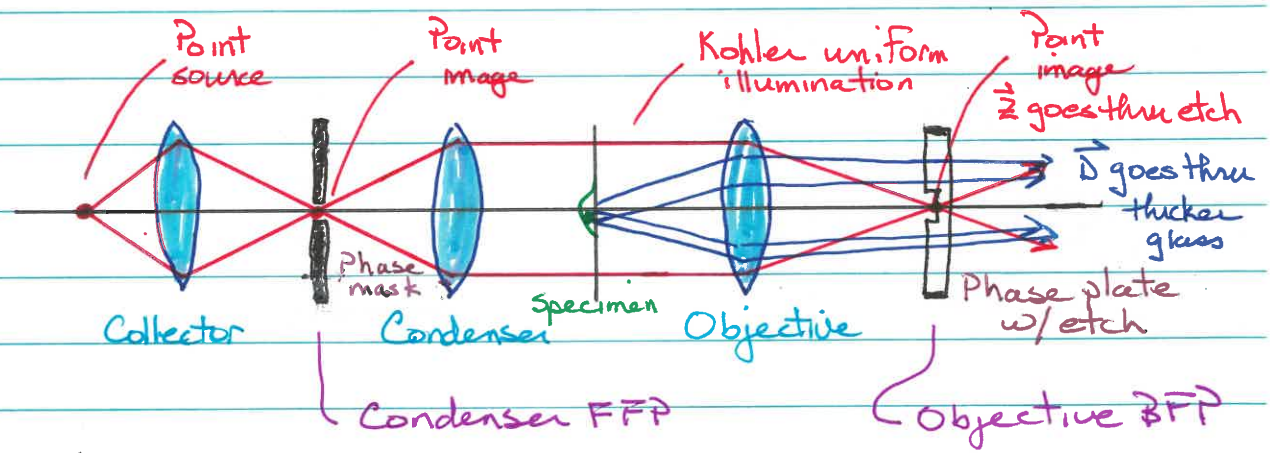
The proper etch thickness will advance the phase by 90°
Calculate the "proper" thickness in this week's homework

The $\vec{z}$ wave is directed thru the etch using a "trick" that exploits the fact that the light source is in focus in different planes in the light path

Light source
Condenser FFP
Objective BFP

Successive uscope planes in which the light source is simultaneously in focus

Consider, for the purpose of understanding the mechanism underlying phase contrast, that the light source is a <u>pointsource</u> centered on the optical axis



The light source will be in focus in the FFP of the condenser <u>AND</u> in the BFP of the objective

Centering the etch in the BFP of the objective will cause all illumination (undiffracted) light from the point source to go thru the etch & be phase advanced (by 90° if the etch has the proper depth)

<u>Big Idea</u>: Restrict $\vec{z}$ wave in space so it can be selectively altered (i.e., phase advanced)

Does Zernike want to affect the $\vec{D}$ wave?
NO. The $\vec{D}$ wave is sample information.

The $\vec{D}$ wave is diffracted light. It goes all over the place in the BFP of the objective. Consequently, most of it will miss the etch and go thru the thicker glass. It will NOT be advanced.

Halo artifact. A small amount of diffracted $\vec{D}$ wave light will go thru the etch & be phase advanced. This causes an artifact in phase contrast images that is called a "halo artifact" because it looks like a bright halo around parts of the specimen (especially cell bodies). See PowerPoint image.

Summary
- $\vec{D}$ wave mostly misses the etch & is NOT advanced
- $\vec{Z}$ wave hits the etch & IS advanced

Implementation Pt 2 – Extended light source

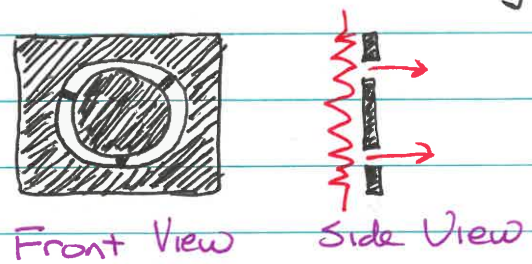
Most light sources are NOT point sources

How do you implement phase contrast w/ an extended source?

Extended light source is also simultaneously in focus in the FFP of the condenser & the BFP of the objective

Illumination can be restricted by putting a mask in the FFP of the condenser (like is done in darkfield μscopy)

Use an annular (ring-like) mask: "phase annulus"



Illumination ($\vec{z}$ wave) light can only pass thru annulus

There are two big advantages to using off-axis annular light instead of an on-axis point source of light:

- Brighter illumination w/ annular light
- Better resolution using off-axis light (see Chapter 4 for explanation)

If the illumination light is an annulus in the FFP of condenser then an annular image of the light will be present in the BFP of the objective because the two planes are mirrors of one another

Implement phase contrast by putting an annular etch in the BFP of the objective that passes only the $\vec{z}$ wave

- Alter illumination in FFP
- Map onto BFP to selectively advance phase only of illumination

$\boxed{\text{ANIMATION}}$ Mol Exp Phase Plate/Ring Alignment

Used animation to explore how corresponding annular structures in condenser FFP & objective BFP are aligned & how, once aligned, they markedly improve contrast

Can look in the BFP on a μscope using a special lens called a "Bertrand lens"